Apache Pig

Employee Salary and Department Analysis

Name: Kotakonda Akshitha

USN: 4GW23CI022

Semester: 7th

1. Aim

To implement Apache Pig for processing and analyzing employee data stored in HDFS, and to calculate the number of employees and average salary for each department.

2. Problem Statement

An organization maintains a large amount of employee information containing employee ID, name, department, age, and salary. When the number of employees becomes large, manually analyzing this information becomes difficult.

The employee data is stored in the **Hadoop Distributed File System (HDFS)**. Apache Pig is used to process the data efficiently.

The Pig program will:

1. Load employee data from HDFS.
2. Filter valid employee records.
3. Group employees according to their department.
4. Calculate the number of employees in each department.
5. Calculate the average salary of each department.
6. Display and store the final result.

The processed data can also be used as a data-preprocessing step in a larger ML/MLOps pipeline.

Tools and Environment Used:

Tools and Environment Used:

Operating System	Ubuntu 22.04 LTS
Java	OpenJDK 11
Hadoop	Apache Hadoop 3.4.3
Data Processing Tool	Apache Pig 0.17.0
Scripting Language	Pig Latin
Storage	HDFS
Processing Engine	Hadoop MapReduce / YARN
Setup	Single-node Hadoop cluster

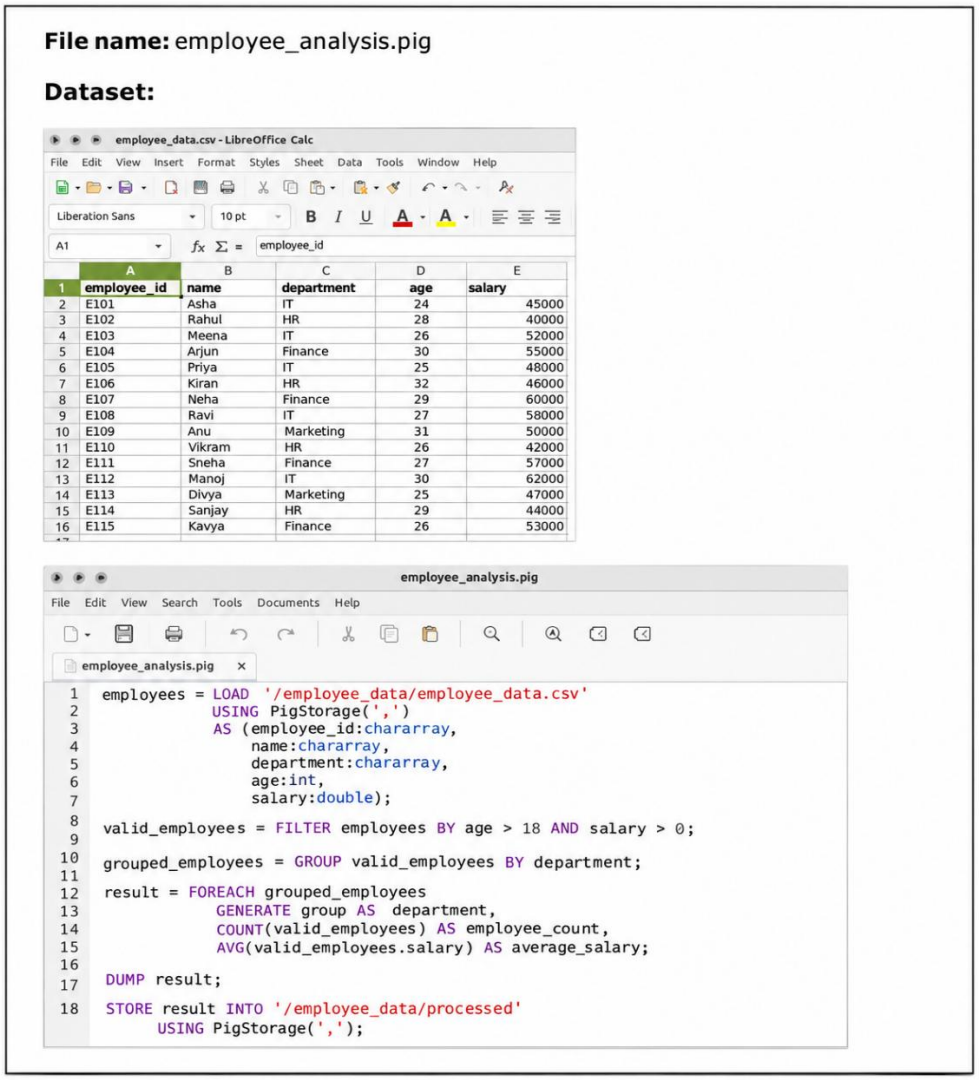
Input Data:

The input file employee_data.csv contains 15 employee records. Each record consists of an Employee ID, Employee Name, Department, Age, and Salary. The dataset includes employees from four departments: IT, HR, Finance, and Marketing. The file is stored in the Hadoop Distributed File System (HDFS) at /employee_data/employee_data.csv. Apache Pig is used to load and filter valid employee records, group them according to their departments, and calculate the total number of employees and average salary for each department. The processed results are then displayed and stored in HDFS.

Code Implementation:

File Name: employee_analysis.pig

DataSet:



employee_id	name	department	age	salary
E101	Asha	IT	24	45000
E102	Rahul	HR	28	40000
E103	Meena	IT	26	52000
E104	Arjun	Finance	30	55000
E105	Priya	IT	25	48000
E106	Kiran	HR	32	46000
E107	Neha	Finance	29	60000
E108	Ravi	IT	27	58000
E109	Anu	Marketing	31	50000
E110	Vikram	HR	26	42000
E111	Sneha	Finance	27	57000
E112	Manoj	IT	30	62000
E113	Divya	Marketing	25	47000
E114	Sanjay	HR	29	44000
E115	Kavya	Finance	26	53000

Execution Steps:

Step 1: Start Hadoop services and check their statusStart HDFS and YARN, then verify that the services are running.

>>start-dfs.sh

>>**start-yarn.sh**

>>jps

Step 2: Create the project directory

>>mkdir ~/employee_pig

>>cd ~/employee_pig

Step 3: Create the input dataset

>>nano employee_data.csv

Step 4: Create an HDFS directory

>>hdfs dfs -mkdir -p /employee_data

Step 5: Upload the dataset to HDFS

>>hdfs dfs -put employee_data.csv /employee_data/

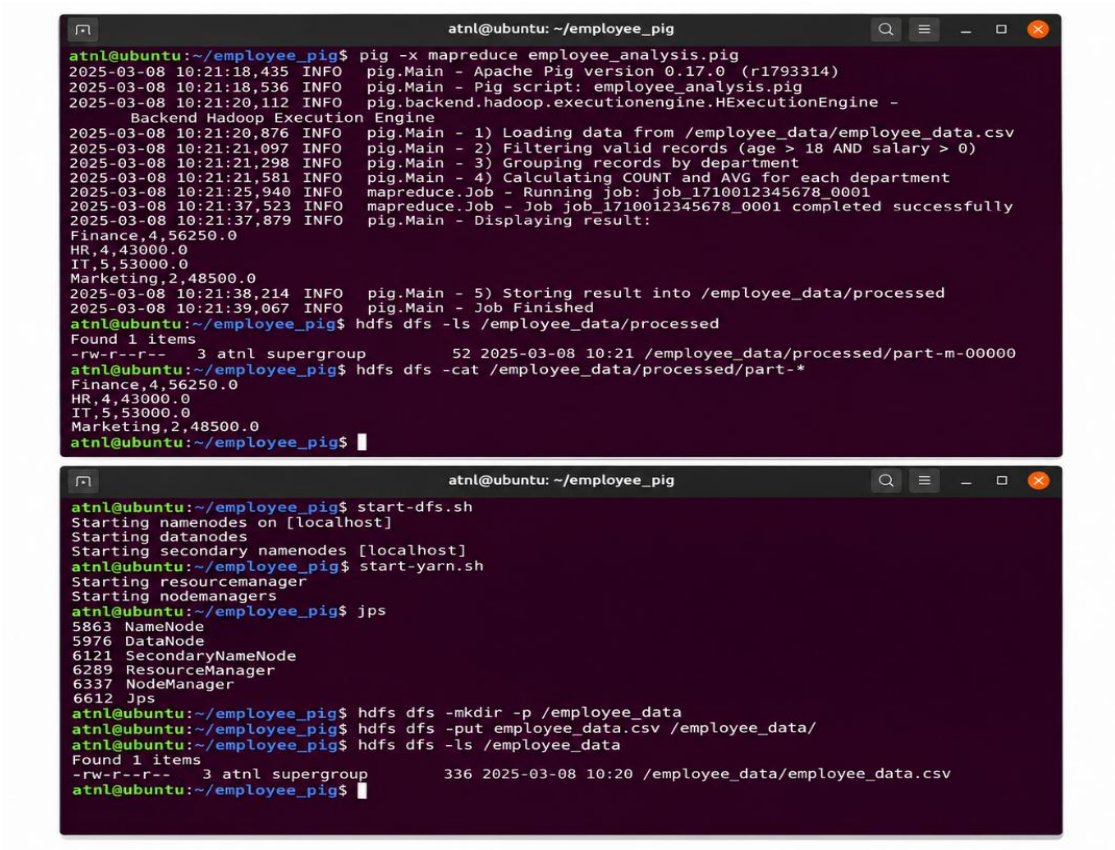
>>hdfs dfs -ls /employee_data

Step 6: Create the Apache Pig program

>>nano employee_analysis.pig

Step 7: Execute the Apache Pig program

>>pig -x mapreduce employee_analysis.pig



Execution Output:

The Apache Pig program completed successfully. The final output was:

Department	Employee Count	Average Salary (₹)
Finance	4	56250.0
HR	4	43000.0
IT	5	53000.0
Marketing	2	48500.0

Brief Discussion

The Apache Pig program processes employee data in several stages. In the LOAD stage, the employee records are read from HDFS using PigStorage(','), which identifies commas as field separators. The FILTER operation selects employees whose age is greater than 18 and whose salary is greater than zero. The GROUP operation groups the valid employee records according to their departments. The FOREACH operation, together with the COUNT and AVG functions, calculates the total number of employees and the average salary for each department. Finally, the DUMP operation displays the results, while the STORE operation saves the processed data back into HDFS.

Department	Grouped Values	Pig Output
Finance	55000,60000,57000,53000	4 employees
HR	40000,46000,42000,44000	4 employees
IT	45000,52000,48000,58000,62000	5 employees
Marketing	50000,47000	2 employees

Conclusion

The Employee Salary and Department Analysis program was implemented using Apache Pig on a Hadoop environment. The employee dataset was stored in HDFS and processed using Pig Latin operations such as LOAD, FILTER, GROUP, FOREACH, COUNT, and AVG. The program calculated the number of employees and average salary for each department and displayed the results. The processed output was stored in HDFS for further use. This practical demonstrates how Apache Pig simplifies large-scale data processing and how the resulting data can support the preprocessing stage of an MLOps pipeline before machine-learning model development.